

Cornell
Engineering

ODP7: Final Report

CHASR Solutions

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User Manual

Candom User Manual

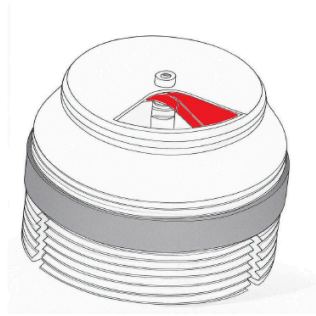
Twist, Lock, Enjoy!

Congratulations on your new Candom!

The Candom prevents spills from canned drinks, allowing you to save your beverage for later and transport it with ease. Its unique tapered thread and locking ring mechanism, combined with a secure O-ring seal, ensure a spillproof fit for a variety of can sizes. This manual is intended as a setup and general use guide to provide a seamless Candom experience.



Step 2: Position the Candom



Step 3: Twist the Locking Ring (highlighted in grey)

General Use

1. Prepare the Can: Ensure the top of your canned drink is clean and free of debris. Open the can if not already opened.
2. Position the Candom: Place the Candom over the top of the can, aligning the drinking holes.
3. Twist the Locking Ring: Hold the Candom firmly and twist the locking ring (highlighted in grey) clockwise. The tapered threads will adjust to the can's diameter as you tighten. Continue twisting until the Candom feels secure and snug. Close the lid (highlighted in red) until you feel a click.
4. Verify the Seal: Gently shake the can to confirm the Candom is locked in place and no liquid escapes. If needed, tighten the locking ring further.

Maintenance

1. Wash the Candom with warm, soapy water after each use. The device is dishwasher-safe for easy cleaning!
2. The Candom is recyclable. That being said, if problems arise, reach out to our support team for replacement and warranty.

Enjoy Your Candom! With its easy twist-and-lock design, the Candom makes saving and not spilling your canned drinks effortless. For further assistance, visit our website at www.candom.com or contact our support team at support@candom.com. Twist, lock, and enjoy your spill-free experience!

[Link to User Manual Doc](#)



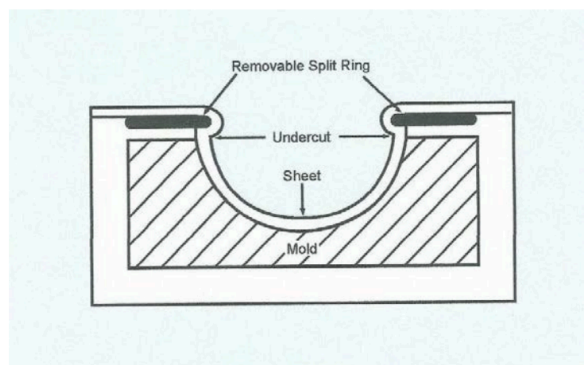
Design For Manufacturing

In order to mass-produce our product, we choose to injection mold the primary body and locking nut parts since they both have high tolerance threads that require smooth movements. The primary has a slightly more complex geometry. Because the lid mechanism is mostly flat we decided on using vacuum forming to create the desired geometry. We choose these manufacturing methods for their high productivity and ability to manufacture our product at scale.

Part	Process	Geometric Freedom	Cost (Tooling + Per Part)	Material Category
Primary	Injection Molding	High: Needs male threads to match locking nut	\$\$	Plastic
Locking Nut	Injection Molding	Medium/High: Needs female threads to match	\$\$	Plastic
Locking Lid Mechanism	Vacuum Forming	Medium: Needs to match the top face of the primary, but mainly simple geometry	\$	Plastic

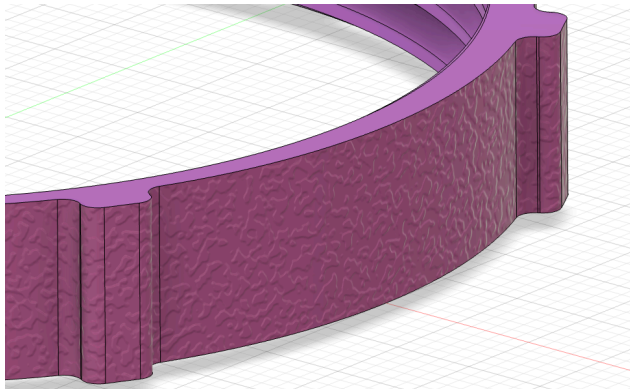
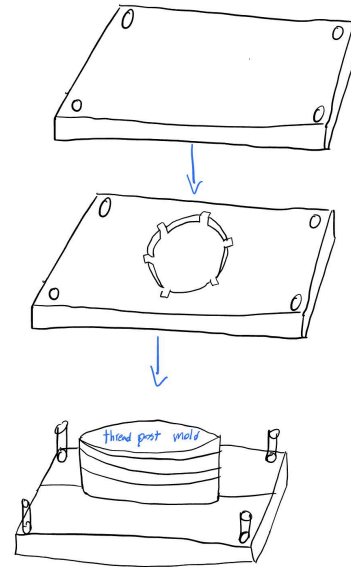
Even though the locking lid mechanism has undercuts it can still be vacuum formed using movable mold sections which can be found in Universal plastics [Thermoforming Design guide](#). This allows for cheaper tooling compared to injection molding while not changing the design.

Diagram of Undercut with moveable mold sections



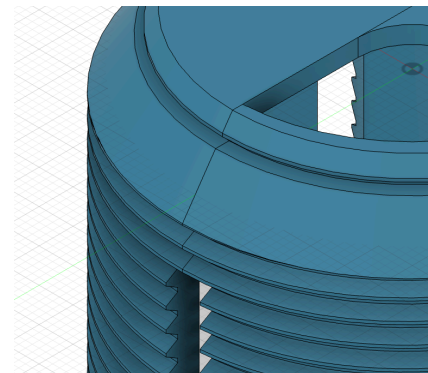


Since the primary and locking nut are both injection molded we expect both parts to include parting lines. For both internally and externally threaded parts there is often a threaded core that gets ejected from the machine or a fully automatic release that unscrews the part from the mold. Since the locking nut is an internally threaded part the parting lines are not going to show up as predominantly as for an externally threaded part. To the right is a tooling mock up which demonstrates that the expected parting lines are going to be along the edges of the part. Since the parting lines are going to be along the parts pre-existing edges the CAD model does not require any updating to show where they would be.



Additionally some changes were implemented to improve the ergonomics of the locking nut. We added grips, and a rough texture to the outside of the nut to allow for easier tightening of the nut. To get a rougher texture the tooling surface can be roughed up via sand blasting or other surface altering processes.

The injection molding for the primary will likely consist of an internal mold piece and two external pieces that form the threads. The interface of the internal and external mold piece will cause parting lines along existing edges. For the interface between the two external mold pieces that will cause a parting line across the top and down along the threads. Parting along threads can cause them to bind more often, so different tooling that limits parting lines along threads such as threaded components could be a future manufacturing improvement.





Design For Assembly

Our product prioritises ease of use and universality above all other characteristics. We did not want to add too many processes for assembling and using the system. An advantage of our design is that it has very simple and large components, making it very easy to assemble and maintain. There are no intricacies that could cause difficulties for the user. One key frustration is the screw used for securing the top of the lid to the rest of the primary body that fits around the can. The screw has resulted in some leakage and complexity that we do not want for our design. It is a redundant reinforcement of the lid while also providing a new exit space for liquid to leak. It is also a harder material than the lid itself, which could cause wear on the lid with prolonged use, which would further increase leakage. The screw also required having a mating component such as a nut or heat set insert which further increased the complexity of the design and difficulty of assembly.

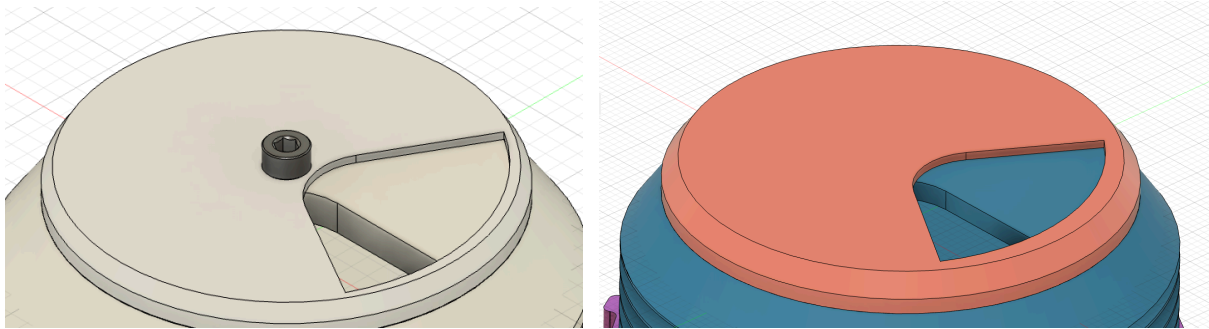
One issue with the existing snap-mechanism of the lid is that it can be challenging to assemble because you need to work the lip of the lid around the primary. Trying to manhandle the lid onto the primary with only two hands is rather difficult. We also found that during initial assembly sometimes the primary and lock nut would not thread on smoothly requiring some breaking in. Having a quality checking station during assembly or specific checking process while the product is being assembled could improve the ease of assembly. This quality check process should be done before the lid is assembled onto the primary that way primaries that fail inspection don't waste lids on it.

The O-ring assembly is pretty easy since there really only is one groove that the O-ring sits nicely in. To assemble it you have to get one side of the O-ring in and hold it in with your finger. Then you have to take your other fingers and push it in along the groove until it seats.

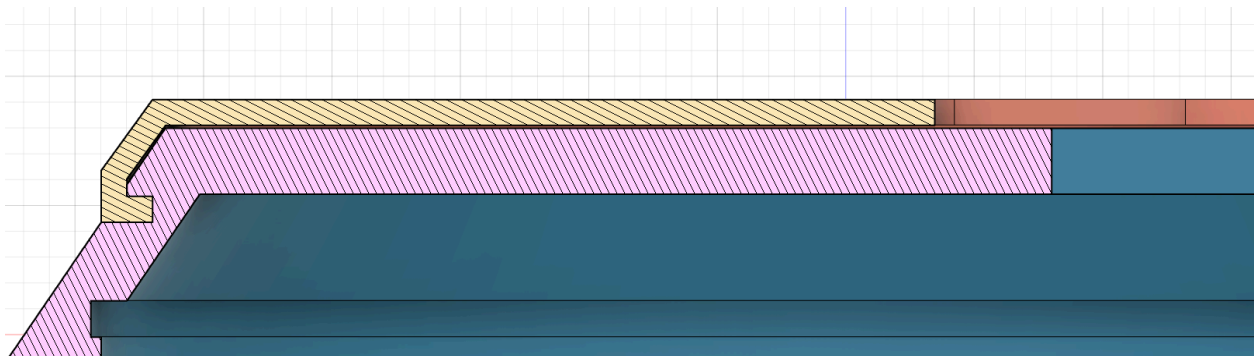


Improved Assembly

The issue of the screw could be solved by developing a mechanism that does not need to puncture the entire lid to allow for the cap to swivel. This would keep the lid entirely closed, minimizing areas for leakage, and would also lessen the complexity of assembly. We discovered that the lid still rotates smoothly even when removing the screw just based on the existing snap on mechanism. By using a snap-mechanism, we eliminate the need for screws and bolts while also allowing for a more tight seal. It also keeps the material consistent throughout the product lessening its assembly complexity while also limiting wear on the plastic material.



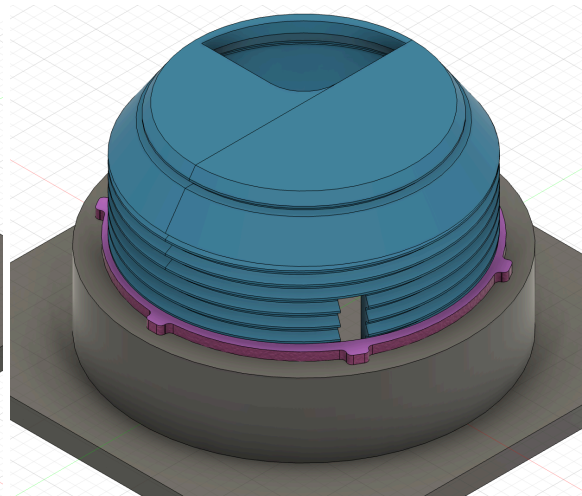
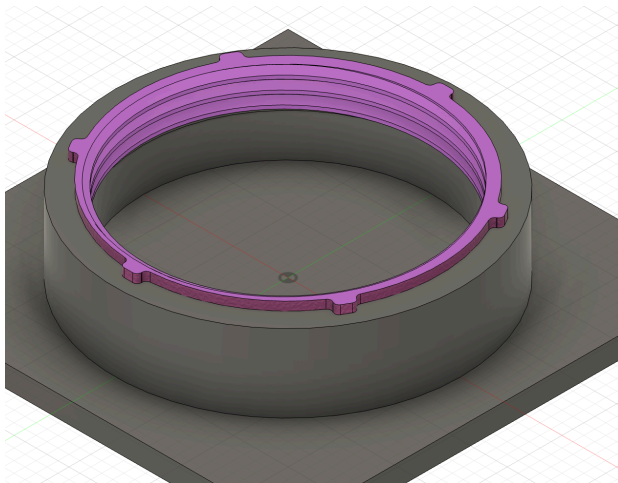
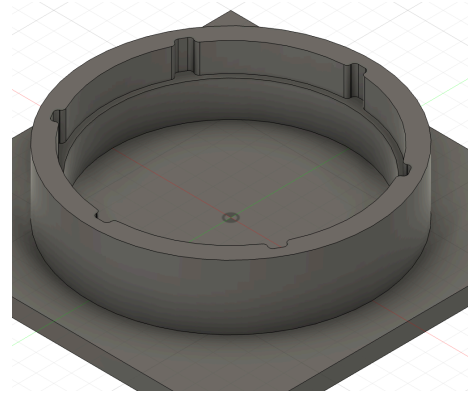
Left: Before screw removal, Right: after screw removal.



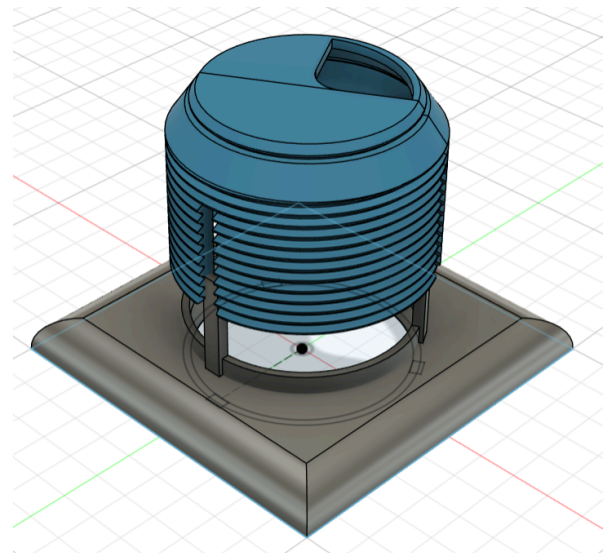
The lid is secured onto the primary solely with a lip that gets snapped into place during assembly.



To improve assembly of attaching the nut to the primary we designed a fixture that holds the nut in place while the primary is twisted on. The height of the fixture walls does not go over the height of the nut in the fixture so that nut can be easily removed if needed. The fixture matches the profile of the nut so a torque can be applied when twisting down. The nut is raised above the base so that the primary can be threaded on a consistent amount.



To improve the easability of assembling the primary to the lid, we developed a fixture that holds the primary down using the three slits already in the primary as fixturing points. A small ridge was added to make it easy to remove the primary from the fixture once it was assembled. A large base was designed to supply sufficient surface area support to minimize any movement that may occur while assembling the components.





Design For Sustainability

Our product is primarily designed to be sustainable by utilizing recyclable materials and a long product lifecycle. We choose to use High-Density Polyethylene (HDPE) for both the primary body and the locking nut because HDPE is considered highly recyclable—being able to be reformed into new products for up to 10 times—and is accepted in most US recycling centers. HDPE is an excellent material to use in injection molding because it has fantastic flow properties which allow for minimized material waste. HDPE parts can be produced in a single-step injection modeled process reducing the need for additional post processing steps. As the world continuously progresses to more sustainable green energy the environmental impact of running injection molding machines will reduce.

We designed our product to have a 10+ year lifecycle by using HDPE for two components since it has excellent corrosion properties, and utilizing TPU for the lid due to its good corrosion resistance. The limiting factor of our product lifecycle is the O-rings since they will last 10-15 years depending on how often they are used.

All materials in the product are recyclable to varying extents; previously mentioned is HDPE, but TPU is recyclable but to a lesser extent than HDPE and O-rings made from Nitrile Rubber can be recyclable but often require specialized recycling services not typical US recycling centers. The tear down to dispose of the parts correctly involved the simple removal of the o-ring and TPU lid. The HDPE parts can be placed in a recycling bin, while the TPU and o-ring need more specialized recycling facilities.

With a total carbon footprint of 0.57 kg CO₂e the most prominent factor impacting our products' carbon footprint is assembly of the processed parts. 58% of the carbon footprint is from the assembly phase consisting of 0.30 kg CO₂e per product. Another 20% (0.1 kg CO₂e) comes from material acquisition to the production plant while processing the material uses 13% or 0.07 kg CO₂e. While shipping from manufacturing plants in China to a distribution center based in the United States is only 3%, and packaging an additional 6% of the total.



O-rings are the most commonly used sealing component in the industrial world, and selecting the right type can make a tremendous difference in system performance, durability, and maintenance. With the increasing concerns about environmental impacts and sustainability, eco-friendly O-ring materials have gained popularity. These materials not only provide excellent sealing performance under harsh conditions but also reduce the carbon footprint of the production process. In this article, we will discuss the benefits of going green with eco-friendly O-ring materials and the available options.

Improved Sustainability

To improve the sustainability of our product, changing the O-ring material from Nitrile Rubber to a more eco-friendly sealing material could improve the recyclability of our product as well as reducing manufacturing impact. Nitrile Rubber often contains harmful chemicals and generates toxic waste during manufacturing and disposal. The Nitrile O-ring could be replaced with an eco-friendly material such as silicon which is made from abundant natural resources. Silicon fulfills material design constraints such as food safety, but can only be recycled in specialized facilities not municipal recycling centers.

Changing the lid material to be HDPE rather than TPU can improve the recyclability of the product. HDPE's sealing ability is considered decent and could maintain the same functionality as the TPU part. With the lid being made from HDPE the entire main body can be recycled at a municipal recycling center, and the O-ring being the only part that needs to be removed from our product before being recycled. This change reduces the cost for consumers to recycle our product significantly since removing the lid is challenging.

By changing the manufacturing location from China to being domestically produced in the United States the carbon footprint can be reduced by 35%. HDPE can be sourced domestically from La Porte, TX—near Houston, TX—where there is a production facility. The manufacturing, assembly and distribution center can be based in Houston so there are no transportation costs between facilities. Material acquisition costs remain the same at 0.1 kg CO₂e—an overall percent increase to 28% from



20%—because the transportation method has to change from ships to trucks. The largest contributing factor is still assembly, but the carbon footprint got reduced to 0.19kg CO₂e from 0.3kg CO₂e. Material processing impact was reduced to 0.05kg CO₂e and packaging went from 3% to 8% but only costs 0.03 kg CO₂e. Additionally, changing from the national grid to renewable energy sources reduces material process costs down to 0.01kg CO₂e lowering the total carbon footprint to 0.34kg CO₂e.

Documentation Package

Zip File & CAD Package includes BOM detailing all purchases for ODP, CAD Package, Prototyping Files: [Google Drive Link to Zip](#).

The CAD package contains:

“Locking Lid Mechanism ODP7.f3z” — Lid of our design;

“Locking Nut ODP7.f3d” — Outer locking nut for our design;

“Primary ODP7.f3z” — Primary body of the product;

“ODP7 Product Assembly.f3z” — Full assembly of the primary, locking nut and lid;

“ODP7 Nut Tightening Fixture.f3z” — Fixture for assembly of the nut onto the primary;

“ODP7 Nut Fixture Assembly.f3z” — Assembly for nut fixture with nut and primary;

“ODP7 Primary Fixture.f3d” — Fixture for assembly of lid onto the primary;

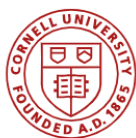
“ODP7 Primary Fixture Assembly.f3z” — Assembly for primary fixture lid attachment;

The Prototyping Files contains:

“Locking Lid Mech ODP7.stl” — 3D printing file for the locking lid mechanism;

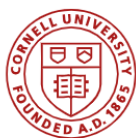
“Locking Nut ODP7.stl” — 3D printing file for the locking nut;

“Primary ODP7.stl” — 3D printing file for the primary body;



Budgeting and Spending

Prototype 1(ODP3)				
Item:	Purchase d/Made	Purpose	Price	Units
Primary	Made	Clamping mechanism that is placed on top of can	11.82	1
Locking Nut	Made	Twisted onto primary allowing for the primarily to be tightened onto the can	3.28	1
Locking Mechanism	Made	A top that spins on the primary, and allows for the can to be closed and not spill	2.93	1
Total Cost of prototype 1			18.03	
Prototype 2(ODP3)				
Primary Lid	Made	Clamping mechanism that is placed on top of can	12.16	1
Locking Nut	Made	Twisted onto primary allowing for the primarily to be tightened onto the can	4.43	1
Tab Lid	Made	A top that spins on the primary, and allows for the can to be closed and not spill	1.05	1
Total Cost of prototype 2			17.64	
Prototype 3(ODP3)				
Locking Nut	Made	A nut twisted onto the primary to fit a 12oz can.	8.31	1
ODP 3 Cost			43.98	
Final Prototype(ODP5)				
Item:	Purchase /Made	Purpose	Price	Units



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Primary	Made	Clamping mechanism that is placed on top of can	18.75	1
Locking Nut	Made	Twisted onto primary allowing for the primary to be tightened onto the can	4.00	1
Locking Mechanism	Made	A top that spins on the primary, and allows for the can to be closed and not spill	2.59	1
O-Rings	Purchase	Sealing the primary against the can (25 pc)	8.30	1
Heatset insert	Free	Screwing a bolt into the primary	0	1
M3 screw	Free	Attaching Locking mech to primary	0	1
Total Cost			33.64`	
Final Cost			77.61	



Photos of Prototypes

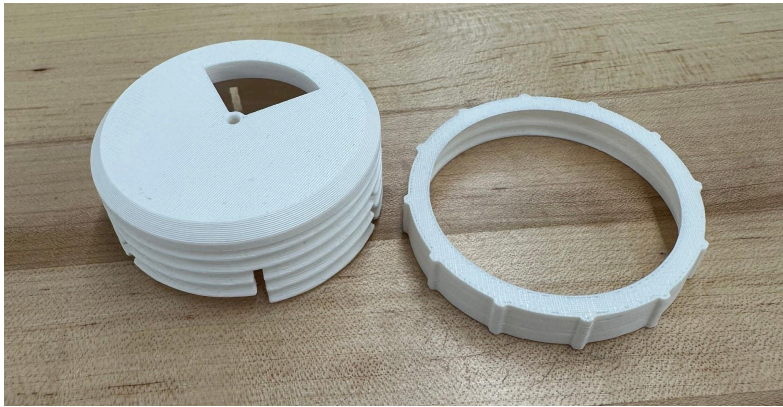


Prototype 1:





Prototype 2:



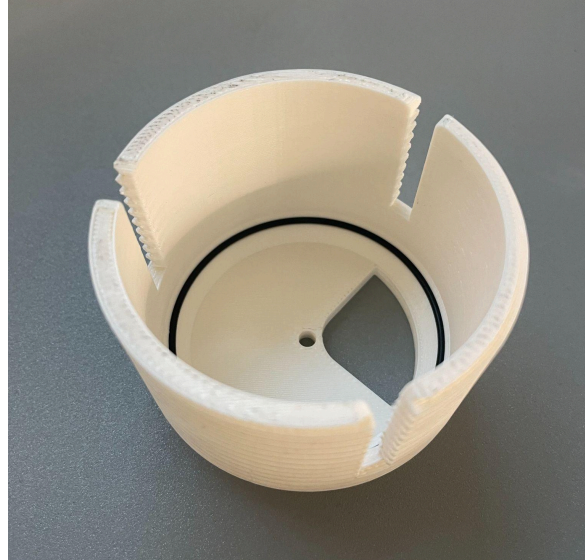
Prototype 3:



Final Prototype:



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Assembly Instructions

1. Assemble the o-ring into the primary casing.
Work the o-ring down into the groove and push along the ring to make sure its entirely in its groove
2. Slide the screw through the hole in the cap
3. Threat the screw (with the cap) into the hole on the top of the primary housing
4. Squeeze along the edge of the lid to help the cap into the groove on the outer side of the primary housing
5. Turn the entire assembly over and screw the bolt slowly onto the threads. Once the bolt has caught the threads, screw it all the way up





Final Recommendation

The Candom project should be discontinued; once we tried implementing the tapered threads (paramount to the key functionality of universality), our product just didn't work as well as we intended. On top of that, an honest assessment and analysis on the market suggest limited feasibility and insufficient interest to justify further investment. While the product aims to solve the problem of spills and storage for canned beverages, the niche it targets—consumers wanting to save and transport opened cans—is relatively small and is not a sustainably viable business. Competitors, such as silicone can lids, resealable toppers, and even reusable bottles, already address similar needs at lower price points or with greater versatility, reducing the Candom's competitive edge. Even then, there's incredibly minimal demand for said products; how many times have you seen a can topper used? The tapered thread and O-ring seal, while innovative, introduce complexity and potential cost barriers that may deter mass-market adoption. In our tests, the importance of a plastic that's both flexible and impact-resistant (essential again for the tapered threads and locking mechanism) further complicates the possibility of manufacturing at scale.

Initial market analysis indicates that consumers prioritize convenience and affordability in beverage accessories and in the beverage experience itself; the Candom's specialized functionality may not command a premium price which is the only way to realistically sustain itself commercially and the convenience we once thought it'd provide.

Extending the development cycle for redesign or market pivoting is unlikely to overcome these challenges, as the core concept faces inherent limitations in scalability and differentiation. Resources would be better allocated to alternative products with broader appeal and stronger market potential, avoiding the sunk costs of pursuing a concept with limited commercial viability.



Reflection

Within our group's design process, we have gone through many steps that allowed each member to find what they enjoyed and didn't enjoy. These steps include brainstorming ideas for what our project should be; preliminary design in sketching and CAD; scrappy prototyping; testing; iterations; and large scale manufacturing planning. Although brainstorming ideas seems like an easy process, it was quite difficult for us as a group to find something that we were all passionate about, as well as something that fit the guidelines in our project. We generally tended towards ideas that we were all passionate about, but either weren't feasible or weren't within the guidelines of the assignments.

The next process is probably where our group showed our strengths with developing preliminary ideas and designs. This was the first time we split up the work within our team based on our strengths, which later allowed for a similar structure for the rest of the project. This structure primarily focused on Hugo creating sketches that Sean would develop in CAD, Aaron would create some of the scrappy prototypes, and Ryan and I would record all the information in the ODP process. The team of course didn't strictly stay on this structure for the whole project, with us splitting up roles in some of creating a prototype each to then reflect and see what to move forward with. The idea of our team having a general idea of what people were good at, but also being flexible in what people do for each ODP assignment worked out very well. It allowed us to learn more skills, find more interests, and there were many scenarios where someone got swapped with other work, and someone else in the group would help out. This was generally fine because we were often good with being transparent in our availability.

The most difficult part of the project was probably the iteration process. Because our product is primarily customized and had to be manufactured in house, the iteration process was a lot of changes in design and waiting until that prototype was manufactured and then testing it. The process of multiple people within the group creating a prototype and then merging these prototypes together was quite useful in the process, however when we started combining designs, some parts of the CAD would



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often break in CAD and another problem would arise when we thought we were finding a solution.

Overall, the design, manufacturing, and iterating process allowed each individual within the group to find which part of the process they were good at and which they were the most interested in. This allowed us to work well together as a team to create a final design and format our project into a presentable product that looks into the consumer interface, mass manufacturing strategies, and sustainability influences.

Portfolio Links

<https://cornell-mae-ug.github.io/spring-2025-portfolio-SeanKaiDempsey/>